Linear Optimization

Mathematics 464 Syllabus

Instructor: Patrick Gambill

Meeting: Wilson-Short Hall 13 (Pullman) VECS 125 (Vancouver), TU 10:35 AM - 11:50 AM, Spring 2026

Instructor Email: patrick.gambill@wsu.edu

Office Hours: M 3:00 PM - 4:00 PM, T 3:00 PM - 4:00 PM, F 3:00 PM - 4:00 PM Zoom

Prerequisites: Math 273, some linear algebra and proof writing

We will not have a required text for this course. However, a recommended text and some supplemental texts are listed below. It is a good idea to purchase a copy of the recommended text as the lectures will loosely follow chapters 1-4 of this text. The supplemental texts will have different explanations or additional topics not in the recommended text.

Recommended Text: *Introduction to Linear Optimization,* by Bertsimas and Tsitsiklis

Supplemental Texts:

- *Introduction to Linear Optimization*, by Nemirovski
- *Linear Programming and its Applications*, by Eiselt and Sandblom
- *Operations Research Applications and Algorithms*, by Winston
- *Join Geometries*, by Prenowitz and Jantosciak

Course Goals and Objectives:

In this course, we will study linear optimization. We will study the geometry of polytopes and how this corresponds to linear optimization problems. We will study some of the algorithms used to solve linear programs. We also will spend some time formulating linear optimization problems

and applying the theory learned earlier in class. At the end of this course, students should be able to formulate a wide class of real world problems as linear optimization problems and students should be able to solve these problems using Python.

Grade Criteria

This is the tentative grade makeup of the course. This is subject to change.

- Homework 70%
- Project 25%
- Participation 5%

Homework

We will have a homework assignment every week or two. Each assignment will consist of a combination of proof and computation problems. As homework will be challenging, you are encouraged to work together on these problems. If you collaborate, then you must each submit your own written solutions to the problems and note which students you worked with. You are welcome to use resources besides the book or lecture notes when working through a problem, but you must cite these resources. These policies are in place to avoid plagiarism. Late work will not be accepted unless you have a legitimate excuse and contact me before the due date.

Corrections

Developing proof writing skills is an iterative process. You will make mistakes and learn from feedback. As such, students are encouraged to make corrections on each homework. Corrections submitted within a week of when feedback is provided will be awarded the missed points from the original submission. Corrections will not replace unexcused or missing assignments.

LLM Policy

The use of LLMs is allowed, but discouraged in this course. If you use LLMs to help with coding, you are responsible for the output of the code. If you use LLMs to help with any writing assignments, I reserve the right to ask additional questions about your writing.

Participation

Students will receive full participation points as long as they consistently contribute to in-class activities and discussions.

Project

The final project will involve modeling and solving a real world problem using the methods discussed in class. This project is fairly open ended and will include a report as well as code. Many of you are in the process of searching for a job or applying to graduate school. The final project should help demonstrate a valuable skill to future employers and graduate schools.

Attendance

It is important to attend class, since there will be topics covered in the lecture that are not in the recommended text. If you have to miss class, you are responsible for getting notes from a classmate or coming to office hours to cover any missed materials.

Grade Scale

The tentative grade scale is provided below:

Percentage	Guaranteed Grade
[94, 100]	A
[90,94)	A-
[87,90)	B+
[84,87)	B
[80,84)	B-
[77,80)	C+
[74,77)	C
[70,74)	C-
[67,70)	D+
[60,67)	D
[0,60)	F

I may adjust these cutoffs, but only if such a change would benefit students.

Tentative Schedule

The Week Of:	Topics	Additional Notes
Jan. 12th	Syllabus, Motivation, Linear Algebra Review	
Jan. 19th	Examples of LPs, LP Formulation, Conversion to Standard Form	HW 1 Due
Jan. 26nd	Convex and PL functions, Graphical Solutions	HW 2 Due
Feb. 2nd	Geometry of Polyhedra, Convex Geometry, BFS	HW 3 Due
Feb. 9th	Standard Form of Polyhedra, Extreme Points	HW 4 Due
Feb. 16th	Simplex Method	HW 5 Due
Feb. 23rd	Bland's Rule, Big M method	HW 6 Due
Mar. 2nd	Tableau and Revised Simplex Method	HW 7 Due
Mar. 9th	Simplex Method	HW 8 Due
Mar. 16th	Spring Break	
Mar. 23rd	Duality	HW 9 Due
Mar. 30th	Dual Theorem, Dual Simplex Method	HW 10 Due
Apr. 6th	Farkas' Lemma	HW 11 Due
Apr. 13th	Interior Point Methods	HW 12 Due
Apr. 20th	Interior Point Methods	
Apr. 27th	Other Topics (Class Preference)	
May 4th	Finals Week, No Class	Project Due

Academic Integrity Statement

All members of the university community share responsibility for maintaining and promoting the principles of integrity in all activities, including academic integrity and honest scholarship. Students are responsible for understanding the full Academic Integrity Statement. Students who violate WSU's Academic Integrity Policy (identified in WAC 504-26-010(3) and -404) will fail the course, will not have the option to withdraw from the course pending an appeal, and will be reported to the Center for Community Standards. Copying another student's work or copying

from a website is plagiarism. If you have any questions about what is and is not allowed in this course, ask me, and I will happily clarify.

WSU Reasonable Accommodation Statement

Reasonable accommodations are available for students with documented disabilities or chronic medical or psychological conditions. If you have such a condition and need accommodations to fully participate in this class, please visit your campus' Access Center/Services website to follow published procedures to request accommodations. Students may also contact their campus offices to schedule an appointment with an Access Advisor. All disability related accommodations are to be approved through the Access Center/Services on your campus. It is a university expectation that students connect with instructors (via email, Zoom, or in person) to discuss logistics within two weeks after they have officially requested their accommodations. For more information, contact an Access Advisor on your home campus:

- Pullman, WSU Global Campus, Everett, Bremerton, and Puyallup: 509-335-3417 Access Center (https://www.accesscenter.wsu.edu) or email at access.center@wsu.edu.

Lauren's Promise: WSU's Commitment to Addressing Discrimination and Harassment

Lauren McCluskey, a 21-year-old honors student athlete, was murdered on October 22, 2018, on the University of Utah campus by a man she briefly dated. WSU will listen and facilitate support and reporting options if someone is threatening you.

Discrimination, discriminatory harassment, sexual harassment, and sexual misconduct (including stalking, intimate partner violence, and sexual violence) are prohibited at WSU (see Executive Policy 15 – WSU Policy Prohibiting Discrimination and Harassment, the WSU Standards of Conduct for Students, and relevant employee manuals). WSU has instituted procedures to respond to violations of these laws and standards, programs aimed at the prevention of such conduct, and intervention on behalf of victims.

If you are in immediate danger, call 911.

If you are experiencing sexual assault, domestic violence, stalking, discrimination or harassment, you have support and options. If you share information with your course instructor, please know that they are required to reach out to the Title IX Coordinator in WSU's Office of Compliance and Civil Rights (CCR), and CCR will reach out to you with information about on and off campus reporting options and resources. CCR is a system-wide resource (all campuses) which is available for intake consultations for you to learn more about available support. You can reach them directly at 509-335-8288, ccr@wsu.edu, or report online (anonymous reports accepted).

You can also speak to a victim advocate, a medical provider, or counselor confidentially about your concerns. Advocates help survivors of crime determine their own needs in regards to their physical and emotional health, reporting options, and academic concerns. At no cost, advocates connect survivors to campus and community services, and provide accompaniment to important appointments (court, hospital, and police) and support throughout the process. For a list of confidential victim advocates and medical providers, please visit CCR Resources.

WSU Police Department (WSU PD) officers and campus security will treat victims of sexual assault, domestic violence, stalking, hate crimes, and other crimes with respect and dignity. WSU PD, campus security departments, CCR, and victim advocates can also help you with safety planning.

Religious Accommodations

Required because Substitute Senate Bill 5166 was signed into Washington State law, effective July 28, 2019. This law requires post-secondary institutions to provide reasonable accommodations for religious observances or activities.

What to include:

Washington State University reasonably accommodates absences allowing for students to take holidays for reasons of faith or conscience or organized activities conducted under the auspices of a religious denomination, church, or religious organization. Reasonable accommodation requires the student to coordinate with the instructor on scheduling examinations or other activities necessary for course completion. Students requesting accommodation must provide

written notification within the first two weeks of the beginning of the course and include specific dates for absences. Approved accommodations for absences will not adversely impact student grades. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the period of absence. Students who feel they have been treated unfairly in terms of this accommodation may refer to Academic Regulation 104 – Academic Complaint Procedures. See also Academic Regulation 82, available at https://registrar.wsu.edu/academic-regulations/.

Safety and Emergency Notification

Classroom and campus safety are of paramount importance at Washington State University, and are the shared responsibility of the entire campus population. WSU urges students to follow the "Alert, Assess, Act," protocol for all types of emergencies and the "Run, Hide, Fight" response for an active shooter incident. Remain ALERT (through direct observation or emergency notification), ASSESS your specific situation, and ACT in the most appropriate way to assure your own safety (and the safety of others if you are able).

Please sign up for emergency alerts on your account at MyWSU. For more information on this subject, campus safety, and related topics, please view the FBI's Run, Hide, Fight video and visit the WSU safety portal.

Full details can be found at https://provost.wsu.edu/classroom-safety/.